

Critical Very–High Cutoffs and Cofactor Occupancy at a Fixed Shift: Cutoff–Band Reciprocity, Sharp Projection Minimax, and Dyadic Localization

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Abstract

The native-affine interface at a fixed shift isolates a very-high-prime face by imposing a cutoff $p > y$. Choosing $y = X^{267/400+\eta}$ with fixed $\eta > 0$ forces its target cofactor into $r \leq X^{133/400-\eta+o(1)}$, apparently separating it from the inherited $R \asymp J = X^{133/400+o(1)}$ packet. We prove that this positive-power displacement is a feature of that convenient cutoff, not a requirement of the literal very-high decomposition. The exact cutoff hypotheses already admit

$$y = C_{\text{vh}}Q, \quad Q = X^{267/400+o(1)},$$

with a fixed support-dependent constant. Writing $\omega_X = y/Q$, the safe and maximal cofactor scales are both $R \asymp J/\omega_X$. Thus a critical cutoff $\omega_X \asymp 1$ preserves the $R \asymp J$ exponent, a subpower cutoff shift has zero fixed-power cost, and only $\omega_X = X^{\eta+o(1)}$ creates the polynomial gap.

Geometric compatibility does not imply energy occupancy. For the literal pre-terminal lift, whose coefficient has no cofactor dependence, we compute the optimal coefficient-uniform projection constant exactly:

$$\beta_{\mathcal{R}} = \min_{m: B_m N(m) > 0} \frac{N_{\mathcal{R}}(m)}{N(m)}.$$

It can vanish under the inherited hypotheses. We give the analogous exact terminal-activation minimax, a largest-prime census of the very-high face, and an exact dyadic decomposition. Conditional on nonzero very-high atomic energy, one compatible dyadic block retains an $X^{-o(1)}$ fraction, so block localization has no positive polynomial loss. It does not prove parent-to-face occupancy. The exact Walsh decomposition also exposes high mass outside the represented terminal image; restoring it is a separate nonmonotone reassembly gate. Finally, under the declared $\Delta_M = O(Q)$ support information and among certified fixed-power schedules, we show that the critical cutoff is support-optimal, while all current support-only multiplicity and automatic-singleton bounds remain far outside the 1/400 endpoint budget. These are exact L0 results and an L1 fixed-shift interface advance; no parity improvement or twin-prime conclusion is claimed.

Keywords: very-high prime cutoff; cofactor occupancy; fixed physical shift; dyadic localization; sharp minimax; parity boundary.

Convention. One nonzero physical shift h_0 , one direct residue/orbit slice, and one side of the inherited terminal system are fixed unless a left/right direct sum is displayed. All atomic norms use raw counting measure. Carriers are fixed independently of the bounded row coefficient vector. The shaped terminal weight $\Lambda_X(n) = (\log n)\Phi_0(n/X)$ is not the von Mangoldt function.

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1 Introduction

The preceding native-affine analysis exposed an upstream support question before either of its two Gram gates can be useful. On the very-high face, one writes

$$n = mj + h_0 = pr, \quad p > y, \quad (1)$$

where p is prime and r is the target cofactor. If one inserts the convenient fixed-power cutoff

$$y = X^{267/400+\eta}, \quad \eta > 0, \quad (2)$$

then $n \leq X^{1+o(1)}$ gives

$$r \leq X^{133/400-\eta+o(1)}. \quad (3)$$

This is disjoint from a dyadic block $R \asymp J = X^{133/400+o(1)}$. TPC-58 therefore recorded a separate parent-to-compatible-band loss λ_{reloc} [6].

That conclusion is correct on the face declared in (2), but it leaves a prior optimization question: is a fixed η forced by the unique-prime dictionary? The answer is no. TPC-45 uses only the literal inequalities

$$y > \max\{|h_0|, D_+, J_+, \Delta_M, \sqrt{N_+}\}, \quad (4)$$

$$\frac{D_+ N_+}{y} < D_- N_-. \quad (5)$$

At the endpoint, $\Delta_M = O(Q)$, every other growing term in the maximum is $o(Q)$, and the second inequality only imposes a bounded lower threshold on y . A sufficiently large fixed multiple of Q therefore satisfies both conditions. The corresponding cofactor ceiling is again a fixed multiple of J .

This cutoff retuning removes a geometric positive-power loss, but it does not move any atom and does not prove that the physical carrier occupies the newly compatible window. The distinction is central:

cutoff compatibility $\neq$ cofactor occupancy $\neq$ terminal activation.

(6)

The first is a support theorem. The second is an orthogonal projection of the pre-terminal carrier. The third imposes the prime equation and all terminal masks. They have different sharp constants and can fail independently.

1.1 Main results

Our first theorem introduces the dimensionless cutoff ratio

$$\omega_X := \frac{y}{Q} \quad (7)$$

and proves the exact reciprocity

$$R_{\text{safe}} = \frac{N_-}{2y} \asymp \frac{J}{\omega_X}, \quad R_{\text{max}} = \frac{N_+}{y} \asymp \frac{J}{\omega_X}. \quad (8)$$

A dyadic block below R_{safe} is automatically contained in the very-high face whenever it terminalizes, while a block above R_{max} is empty. Only $O(1)$ boundary shells lie between these two scales.

Our second result makes occupancy exact. In the literal lift,

$$[G_X^L]_{m,r,j,\gamma} = c_{m,j,\gamma} \zeta_m, \quad (9)$$

with no r -dependence. For a fixed cofactor set $\mathcal{R}$, let $N(m)$ and $N_{\mathcal{R}}(m)$ be the total and retained cofactor multiplicities in row m , and put

$$B_m := \sum_{j,\gamma} |c_{m,j,\gamma}|^2. \quad (10)$$

Then the sharp uniform energy-transfer constant is

$$\beta_{\mathcal{R}} = \inf_{\mathcal{E}_{\text{par}}(\zeta) > 0} \frac{\mathcal{E}_{\mathcal{R}}(\zeta)}{\mathcal{E}_{\text{par}}(\zeta)} = \min_{m: B_m N(m) > 0} \frac{N_{\mathcal{R}}(m)}{N(m)}. \quad (11)$$

This is a rowwise minimax law, not an asymptotic estimate. The inherited carrier assumptions allow $N_{\mathcal{R}}(m) = 0$ on an active row, so $\beta_{\mathcal{R}} = 0$ is possible.

Our third result identifies the terminal face exactly using the largest prime factor of $mj + h_0$, and partitions its raw atomic energy into dyadic cofactor blocks. If that face has positive energy, a maximal block retains an $X^{-o(1)}$ fraction. This proves $\lambda_{\text{blk}} = 0$ after high-face activation. It does not prove any parent-to-face energy transfer, nor does it permit a restricted coherent norm to be returned monotonically to the undeleted output.

1.2 Claim boundary and organization

The cutoff theorem, projection minimax, terminal census, and dyadic partition are exact finite-dimensional or support statements. Their specialization to the literal fixed- h_0 packet is L1. A positive cofactor-occupancy ratio, a positive terminal activation ratio, weighted singleton occupancy, either represented native-affine Gram constant, missing-high/off-carrier reassembly, boundary/tail reconnection, and a lower bound for the original arithmetic defect remain open. Nothing here crosses the classical parity boundary [1].

Section 2 proves cutoff-cofactor reciprocity. Section 3 gives the exact projection and terminal minimax laws. Section 4 proves the largest-prime census and dyadic localization. Section 5 connects the selected block to the undeleted singleton bridge without assuming norm monotonicity. Sections 6 and 7 give the endpoint audit, kill tests, and revised route.

2 Critical cutoff admissibility and cutoff-cofactor reciprocity

We first separate the literal cutoff inequalities from one convenient power-scale choice. This distinction is what permits the endpoint support door to be reopened without changing the physical shift or the coefficient packet.

2.1 Endpoint support data

Let the positive target and source supports satisfy

$$D_- \leq d_m \leq D_+, \quad M_- \leq m \leq M_+, \quad J_- \leq j \leq J_+, \quad N_- \leq n_{m,j} := mj + h_0 \leq N_+, \quad (12)$$

with fixed support-ratio bounds

$$D_{\pm} \asymp D, \quad M_{\pm} \asymp Q, \quad J_{\pm} \asymp J, \quad N_{\pm} \asymp X, \quad \Delta_M := M_+ - M_- \ll Q. \quad (13)$$

At the endpoint,

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad D = X^{10049/52500+o(1)}, \quad QJ \asymp X. \quad (14)$$

In particular,

$$|h_0| + D_+ + J_+ + \sqrt{N_+} = o(Q). \quad (15)$$

We retain the inherited literal one-side row system and assume that each retained source row has its declared unique representation

$$m = d_m \ell_m \quad (16)$$

inside that system. This is the row-factorization hypothesis used when a fixed- (j, p) incidence is lifted back from the integer m to the full source coordinate.

Call a real cutoff $y = y_X$ *very-high admissible* if

$$y > \max\{|h_0|, D_+, J_+, \Delta_M, \sqrt{N_+}\}, \quad (17)$$

$$\frac{D_+ N_+}{y} < D_- N_-. \quad (18)$$

These are the exact TPC-45 hypotheses; no exponent has yet been assigned to the amount by which y exceeds Q [2].

Theorem 2.1 (Critical very-high cutoff). *Under (12)–(15) and (16), there is a fixed constant $C_{\text{vh}} > 0$, depending only on the support-ratio constants, such that*

$$\boxed{y = C_{\text{vh}} Q} \quad (19)$$

is very-high admissible for all sufficiently large X . Consequently the unique very-high-prime factorization, the residual identity $s = d_m r$, and fixed- (j, p) row singleton incidence all hold already at the critical exponent 267/400.

Proof. Choose C_{vh} strictly larger than the fixed implied constant in $\Delta_M \ll Q$. Equation (15) then proves (17) for large X . The ratio

$$\frac{D_+ N_+}{D_- N_-}$$

is bounded by (13), whereas $C_{\text{vh}} Q \rightarrow \infty$. Hence $y > D_+ N_+ / (D_- N_-)$, which is equivalent to (18).

For completeness, $p > D_+$ prevents a very-high prime from lying in d_m , and $n_{m,j} < y^2$ permits at most one prime divisor $p > y$ of $n_{m,j}$. If $n_{m,j} = pr$, the fused label is $d_m n_{m,j} = p(d_m r)$, so the residual label is $s = d_m r$. Finally, if the same (j, p) meets two rows, then $p \mid (m - m')j$. The alternative $p \mid j$ would imply $p \mid h_0$, impossible because $p > |h_0|$. Thus $p \mid m - m'$, but $|m - m'| \leq \Delta_M < p$, so $m = m'$. The unique representation (16) then recovers the same full source row. $\square$

Remark 2.2 (What the theorem does not say). The constant C_{vh} need not place a previously chosen dyadic cofactor block inside the very-high face. It proves that blocks of exponent 133/400 are geometrically compatible after a fixed-factor retuning. A positive atomic-energy intersection is a separate occupancy statement.

2.2 Exact reciprocity for a general cutoff

Put

$$\omega_X := \frac{y}{Q}, \quad R_{\text{safe}}(y) := \frac{N_-}{2y}, \quad R_{\text{max}}(y) := \frac{N_+}{y}. \quad (20)$$

The names refer to dyadic cofactor blocks $[R, 2R)$.

Theorem 2.3 (Cutoff-cofactor reciprocity). *Let y be very-high admissible, and let a terminal atom satisfy*

$$n = mj + h_0 = pr, \quad N_- \leq n \leq N_+. \quad (21)$$

For a dyadic block $R \leq r < 2R$, one has

$$\frac{N_-}{2R} < p \leq \frac{N_+}{R}. \quad (22)$$

Consequently:

- (i) if $R < R_{\text{safe}}(y)$, every terminal prime in the block is automatically greater than y ;
- (ii) if $R > R_{\text{max}}(y)$, the block contains no very-high atom;
- (iii) the interval between $R_{\text{safe}}(y)$ and $R_{\text{max}}(y)$ meets only $O(1)$ dyadic scales.

Moreover,

$$R_{\text{safe}}(y) \asymp R_{\text{max}}(y) \asymp \frac{J}{\omega_X}. \quad (23)$$

All implied constants depend only on the fixed support ratios.

Proof. Since $r < 2R$ and $n \geq N_-$, one has $p = n/r > N_-/(2R)$. Since $r \geq R$ and $n \leq N_+$, one has $p \leq N_+/R$, proving (22). If $R < N_-/(2y)$, the lower endpoint is greater than y . If $R > N_+/y$, then $p > y$ and $r \geq R$ would give $n = pr > yR > N_+$, a contradiction. Finally,

$$\frac{R_{\text{max}}(y)}{R_{\text{safe}}(y)} = 2 \frac{N_+}{N_-} \asymp 1,$$

so only a bounded number of dyadic scales lies between them. Using $N_{\pm} \asymp X \asymp QJ$ and $y = \omega_X Q$ proves (23). $\square$

Corollary 2.4 (Three cutoff regimes). *Suppose $\omega_X = X^{\xi+o(1)}$ with $\xi \geq 0$. Then the top compatible cofactor exponent is*

$$\theta_R = \frac{133}{400} - \xi + o(1). \quad (24)$$

In particular:

- (a) if $\omega_X \asymp 1$, compatible blocks have $R \asymp J$, with no geometrically forced positive-power relocation;
- (b) if $\omega_X = X^{o(1)}$, the displacement is subpower and has zero fixed exponent;
- (c) if $\omega_X = X^{\eta+o(1)}$ for fixed $\eta > 0$, every compatible block satisfies $R \leq X^{133/400-\eta+o(1)}$ and is eventually disjoint from any fixed dyadic block $R \asymp J$.

The last case is precisely the face audited in TPC-58. The first case is not a contradiction; it uses the same literal unique-prime theorem with a sharper admissible cutoff. No coefficient, norm, or physical shift has been changed.

3 Exact cofactor occupancy and terminal activation

The critical cutoff makes a cofactor window possible at the desired exponent. We now determine exactly what is needed for the literal carrier to place energy in that window. This is an orthogonal projection problem before terminal grouping and therefore admits a sharp rowwise solution.

3.1 Literal pre-terminal energy

On one declared source system, let $\mathcal{I}_X$ be the fixed retained set of (m, r) -pairs, $\mathcal{J}_{X,a}$ the fixed orbit class, and $\mathcal{O}_X$ the complete opposite-row cell. The TPC-53/57 factorization can be written

$$[G_X^L]_{m,r,j,\gamma} = c_{m,j,\gamma} \zeta_m, \quad (m, r) \in \mathcal{I}_X, \quad j \in \mathcal{J}_{X,a}, \quad \gamma \in \mathcal{O}_X, \quad (25)$$

where $c_{m,j,\gamma}$ contains the source shape with its Möbius and prime-logarithm factors, literal mask, opposite-row factor, orbit phase, and four-factor profile, but not the bounded row coefficient ζ_m . The essential fact is that $c_{m,j,\gamma}$ has no r -dependence [3, 5].

Put

$$B_m := \sum_{j \in \mathcal{J}_{X,a}} \sum_{\gamma \in \mathcal{O}_X} |c_{m,j,\gamma}|^2, \quad N(m) := \#\{r : (m, r) \in \mathcal{I}_X\}. \quad (26)$$

Then the raw parent energy is exactly

$$\boxed{\mathcal{E}_{\text{par}}(\zeta) = \sum_m B_m |\zeta_m|^2 N(m)}. \quad (27)$$

Repeated cofactors are distinct orthogonal atoms in this identity.

For any fixed cofactor set $\mathcal{R} \subset (0, \infty)$, define

$$N_{\mathcal{R}}(m) := \#\{r : (m, r) \in \mathcal{I}_X, \quad r \in \mathcal{R}\} \quad (28)$$

and the projected energy

$$\boxed{\mathcal{E}_{\mathcal{R}}(\zeta) = \sum_m B_m |\zeta_m|^2 N_{\mathcal{R}}(m)}. \quad (29)$$

The set $\mathcal{R}$ may be the safe critical window, a union of dyadic blocks, or one distinguished block. It is fixed independently of ζ .

Theorem 3.1 (Sharp rowwise occupancy minimax). *Assume $\mathcal{E}_{\text{par}}$ is not identically zero on the coefficient cube $|\zeta_m| \leq 1$. The optimal coefficient-uniform projection constant is*

$$\boxed{\begin{aligned} \beta_{\mathcal{R}} &:= \inf_{\substack{|\zeta_m| \leq 1 \\ \mathcal{E}_{\text{par}}(\zeta) > 0}} \frac{\mathcal{E}_{\mathcal{R}}(\zeta)}{\mathcal{E}_{\text{par}}(\zeta)} \\ &= \min_{m: B_m N(m) > 0} \frac{N_{\mathcal{R}}(m)}{N(m)}. \end{aligned}} \quad (30)$$

In particular,

$$\beta_{\mathcal{R}} > 0 \iff N_{\mathcal{R}}(m) > 0 \quad \text{for every row with } B_m N(m) > 0. \quad (31)$$

The formula is sharp at every finite X .

Proof. For $B_m N(m) > 0$, write

$$\rho_m := \frac{N_{\mathcal{R}}(m)}{N(m)}, \quad w_m := B_m N(m) |\zeta_m|^2.$$

Whenever the denominator is positive,

$$\frac{\mathcal{E}_{\mathcal{R}}(\zeta)}{\mathcal{E}_{\text{par}}(\zeta)} = \frac{\sum_m w_m \rho_m}{\sum_m w_m}, \quad (32)$$

which is a convex combination of the row ratios. It is at least their minimum. Conversely, choose a minimizing row m_* , set $\zeta_{m_*} = 1$, and set every other row coefficient to zero. Then the ratio equals ρ_{m_*} . The positivity test follows immediately. $\square$

For one distinguished coefficient vector, define the exact weaker ratio

$$\rho_{\mathcal{R}}(\zeta) := \frac{\sum_m B_m |\zeta_m|^2 N_{\mathcal{R}}(m)}{\sum_m B_m |\zeta_m|^2 N(m)}, \quad (33)$$

whenever the denominator is positive. Then

$$\beta_{\mathcal{R}} \leq \rho_{\mathcal{R}}(\zeta) \leq 1. \quad (34)$$

A favorable distinguished ratio does not prove a uniform carrier theorem.

Corollary 3.2 (Sharp zero-occupancy obstruction). *If one active row has retained cofactors but none in $\mathcal{R}$, then $\beta_{\mathcal{R}} = 0$. Hence geometric compatibility of $\mathcal{R}$, positive cardinality of $\mathcal{R}$ in other rows, and boundedness of ζ cannot imply a positive uniform parent-to-window transfer.*

This is why the word “relocation” must be handled carefully. The map is the coordinate projection onto already existing r -labels; it does not send an atom at one cofactor to another cofactor. A fixed-factor cutoff retuning can change which existing labels are compatible, but it cannot create their energy.

3.2 Terminal activation as a second rowwise minimax

For a pre-terminal coordinate $u = (m, r, j, \gamma)$, put

$$n_u := mj + h_0, \quad p_u := \frac{n_u}{r} \quad (35)$$

when the quotient is integral. Let $\mathbf{1}_{\text{term}}^{>y}(u)$ be the indicator that u satisfies the complete inherited terminal support: p_u is prime, $p_u > y$, $\Lambda_X(n_u) \neq 0$, and all required squarefree, coprimality, and support conditions hold. Define the rowwise terminal weight inside $\mathcal{R}$ by

$$T_{\mathcal{R}}(m) := \sum_{\substack{r:(m,r) \in \mathcal{I}_X \\ r \in \mathcal{R}}} \sum_{j,\gamma} \mathbf{1}_{\text{term}}^{>y}(m, r, j, \gamma) |\Lambda_X(n_u)|^2 |c_{m,j,\gamma}|^2. \quad (36)$$

The literal terminal amplitude is

$$x_u = \mu(r) \Lambda_X(n_u) c_{m,j,\gamma} \zeta_m, \quad (37)$$

so the raw terminal diagonal in $\mathcal{R}$ is

$$\boxed{\mathcal{D}_{\mathcal{R}}^{\text{term}}(\zeta) = \sum_m T_{\mathcal{R}}(m) |\zeta_m|^2.} \quad (38)$$

Proposition 3.3 (Exact terminal-activation minimax). *Assume $\mathcal{E}_{\mathcal{R}}$ is not identically zero. The optimal uniform lower constant from compatible pre-terminal energy to its very-high terminal diagonal is*

$$\boxed{\tau_{\mathcal{R}} := \inf_{\substack{|\zeta_m| \leq 1 \\ \mathcal{E}_{\mathcal{R}}(\zeta) > 0}} \frac{\mathcal{D}_{\mathcal{R}}^{\text{term}}(\zeta)}{\mathcal{E}_{\mathcal{R}}(\zeta)} = \min_{m: B_m N_{\mathcal{R}}(m) > 0} \frac{T_{\mathcal{R}}(m)}{B_m N_{\mathcal{R}}(m)}.} \quad (39)$$

The exact combined parent-to-terminal constant is

$$\boxed{\chi_{\mathcal{R}} := \inf_{\substack{|\zeta_m| \leq 1 \\ \mathcal{E}_{\text{par}}(\zeta) > 0}} \frac{\mathcal{D}_{\mathcal{R}}^{\text{term}}(\zeta)}{\mathcal{E}_{\text{par}}(\zeta)} = \min_{m: B_m N(m) > 0} \frac{T_{\mathcal{R}}(m)}{B_m N(m)},} \quad (40)$$

with the convention $T_{\mathcal{R}}(m) = 0$ if $N_{\mathcal{R}}(m) = 0$. Moreover,

$$\chi_{\mathcal{R}} \geq \beta_{\mathcal{R}} \tau_{\mathcal{R}}. \quad (41)$$

None of these exact formulas asserts that the displayed constant is positive.

Proof. The two minimax formulas follow from the same weighted-average argument as [Theorem 3.1](#), concentrating ζ on one row for sharpness. If an active parent row has $N_{\mathcal{R}}(m) = 0$, then $\beta_{\mathcal{R}} = \chi_{\mathcal{R}} = 0$, and (41) is immediate. Otherwise every active parent row has $N_{\mathcal{R}}(m) > 0$, and

$$\frac{T_{\mathcal{R}}(m)}{B_m N(m)} = \frac{N_{\mathcal{R}}(m)}{N(m)} \frac{T_{\mathcal{R}}(m)}{B_m N_{\mathcal{R}}(m)}.$$

Taking rowwise minima proves (41). $\square$

The inherited hypotheses impose no prime-distribution lower bound on $T_{\mathcal{R}}(m)$. Thus $\tau_{\mathcal{R}}$ and $\chi_{\mathcal{R}}$ may vanish even when $\beta_{\mathcal{R}} > 0$. Conversely, a positive terminal face in selected rows does not repair a row with zero compatible occupancy. This formally separates the last two gates in (6).

Remark 3.4 (Activation constants are not probabilities). The literal multiplier $|\Lambda_X(n)|^2$ remains inside $T_{\mathcal{R}}(m)$. Consequently $\tau_{\mathcal{R}}$ and $\chi_{\mathcal{R}}$ are energy-transfer constants and can exceed one; only their possible vanishing and their polynomial lower scale are relevant to the loss ledger.

4 Largest-prime census and zero-cost dyadic localization

The preceding minimax constants are exact but still refer to a terminal indicator. We now identify that indicator arithmetically and show that, once very-high terminal energy exists, localizing it to one cofactor block costs only a logarithm.

4.1 The exact largest-prime face

Adopt the convention $P^+(1) = 1$. Let $\mathcal{U}_{h_0}^{\text{term}}$ be the complete literal terminal universe of TPC-57, and define

$$\mathcal{U}_{h_0, > y}^{\text{term}} := \{u = (m, r, j, \gamma) \in \mathcal{U}_{h_0}^{\text{term}} : p_u > y\}. \quad (42)$$

Theorem 4.1 (Largest-prime census). *Assume $y > \sqrt{N_+}$, and fix (m, j, γ) with $n = mj + h_0 \in [N_-, N_+]$. There is at most one terminal atom over this triple with $p > y$. Such an atom exists exactly when*

$$p_* := P^+(n) > y, \quad r_* := \frac{n}{p_*}, \quad (m, r_*) \in \mathcal{I}_X, \quad (43)$$

and (m, r_, j, γ) satisfies the remaining inherited terminal masks, squarefree conditions, coprimality conditions, and $\Lambda_X(n) \neq 0$. In that case*

$$p_u = p_* = P^+(mj + h_0), \quad r_u = \frac{mj + h_0}{P^+(mj + h_0)}. \quad (44)$$

Proof. If two distinct primes greater than y divided n , their product would exceed $y^2 > N_+ \geq n$. Thus a prime divisor above y is unique and necessarily equals $P^+(n)$. The equality $p_u r = mj + h_0 = n$ then forces $r = r_*$. The retained-pair and remaining terminal conditions are plainly necessary. Conversely, if all displayed and inherited conditions hold, p_* is prime by definition of largest prime factor, the equality $p_* r_* = n$ holds, and the corresponding coordinate belongs to $\mathcal{U}_{h_0, > y}^{\text{term}}$. $\square$

For the literal terminal amplitude

$$x_u = \mu(r_u) \Lambda_X(p_u r_u) [G_X^L]_u, \quad (45)$$

define the very-high atomic diagonal

$$\mathcal{D}_H(y; \zeta) := \sum_{u \in \mathcal{U}_{h_0, > y}^{\text{term}}} |x_u|^2. \quad (46)$$

The theorem gives the exact weighted census

$$\mathcal{D}_H(y; \zeta) = \sum_{m,j,\gamma} \mathbf{1}_{\text{vh}}(m, j, \gamma; y) |\Lambda_X(mj + h_0)|^2 |c_{m,j,\gamma}|^2 |\zeta_m|^2, \quad (47)$$

where $\mathbf{1}_{\text{vh}}$ is precisely the test in [Theorem 4.1](#). Under the sign calibration of TPC-58, $x_u = -\mu(d_m r_u) b_{i,d_m r_u p_u}$, so

$$|x_u|^2 = |b_{i,d_m r_u p_u}|^2. \quad (48)$$

Thus every matched terminal/Walsh coordinate has the same raw atomic weight on both sides of the dictionary [6]. This coordinate injection is not asserted to be surjective onto the full Walsh carrier.

The census is an identity, not a density theorem. It permits $\mathcal{D}_H(y; \zeta) = 0$ while the parent or full Walsh diagonal is positive: the targets may all be y -smooth, the required complementary cofactors may be absent from $\mathcal{I}_X$, terminal masks may fail, or the distinguished row coefficients may vanish on every surviving row.

4.2 Exact dyadic partition

Put

$$R_+(y) := \frac{N_+}{y}. \quad (49)$$

If $R_+(y) < 1$, the very-high terminal face is empty. Otherwise, for integers

$$0 \leq k \leq K_X := \lfloor \log_2 R_+(y) \rfloor, \quad (50)$$

define the half-open cofactor cells

$$\mathcal{B}_k(y) := \{u \in \mathcal{U}_{h_0, > y}^{\text{term}} : 2^k \leq r_u < 2^{k+1}\} \quad (51)$$

and their atomic energies

$$\mathcal{D}_k(y; \zeta) := \sum_{u \in \mathcal{B}_k(y)} |x_u|^2. \quad (52)$$

Theorem 4.2 (Exact dyadic localization). *The cells $\mathcal{B}_k(y)$ form a disjoint partition of the very-high terminal face and*

$$\mathcal{D}_H(y; \zeta) = \sum_{k=0}^{K_X} \mathcal{D}_k(y; \zeta). \quad (53)$$

If $\mathcal{D}_H(y; \zeta) > 0$, let k_* be the least index attaining the maximum dyadic energy. Then

$$\mathcal{D}_{k_*}(y; \zeta) \geq \frac{\mathcal{D}_H(y; \zeta)}{B_X(y)}, \quad B_X(y) := 1 + \lfloor \log_2 R_+(y) \rfloor. \quad (54)$$

For every admissible endpoint cutoff $y \geq C_{\text{vh}} Q$,

$$B_X(y) = O(\log X) = X^{o(1)}. \quad (55)$$

Hence dyadic localization inside an already activated very-high terminal face has zero fixed-power loss.

Proof. Every very-high atom has positive integral $r_u \leq N_+/p_u < N_+/y$, so it lies in exactly one half-open cell in the displayed index range. Raw atomic energy is additive across disjoint coordinate sets, proving (53). The maximum of $B_X(y)$ nonnegative numbers is at least their average. Finally, $R_+(y) \leq N_+/(C_{\text{vh}} Q) = X^{133/400+o(1)}$, proving (55). $\square$

In the TPC-58 loss notation, the theorem proves

$$\boxed{\lambda_{\text{blk}} = 0} \quad (56)$$

conditional on a lower bound for $\mathcal{D}_H$. It does not prove a parent-to-high-face energy transfer: the ratio $\mathcal{D}_H/\mathcal{E}_{\text{par}}$, or even $\mathcal{D}_H/\mathcal{D}_0$, remains uncontrolled.

4.3 The base/represented/missing decomposition and its limit

Let $\mathcal{W}_0$ be the complete declared Walsh slice and define its raw atomic diagonal by

$$\mathcal{D}_0 := \sum_{w \in \mathcal{W}_0} |b_w|^2. \quad (57)$$

Let $\mathcal{W}_H(y) \subseteq \mathcal{W}_0$ be the set of high Walsh atoms in that slice, namely those whose unique very-high prime exceeds y , and let $\iota : \mathcal{U}_{h_0, > y}^{\text{term}} \rightarrow \mathcal{W}_H(y)$ be the injective terminal/Walsh coordinate map. Its image retains exactly the retained-pair and terminal-mask conditions in [Theorem 4.1](#). Define

$$\mathcal{D}_{\text{base}}(y) := \sum_{w \in \mathcal{W}_0 \setminus \mathcal{W}_H(y)} |b_w|^2, \quad (58)$$

$$\mathcal{D}_{\text{miss}}(y) := \sum_{w \in \mathcal{W}_H(y) \setminus \text{Ran } \iota} |b_w|^2. \quad (59)$$

The second term is the high Walsh mass whose complementary cofactor is absent from the retained carrier or whose remaining terminal conditions fail. Unique very-high factorization and the atomic norm match (48) now give the exact three-part decomposition

$$\boxed{\mathcal{D}_0 = \mathcal{D}_{\text{base}}(y) + \mathcal{D}_{\text{miss}}(y) + \sum_{k=0}^{K_X} \mathcal{D}_k(y).} \quad (60)$$

In particular, with $\mathcal{D}_{\text{off}}(y) := \mathcal{D}_{\text{base}}(y) + \mathcal{D}_{\text{miss}}(y)$, either

$$\mathcal{D}_{\text{off}}(y) \geq \frac{1}{2} \mathcal{D}_0 \quad (61)$$

or some represented high cofactor block satisfies

$$\mathcal{D}_k(y) \geq \frac{\mathcal{D}_0}{2B_X(y)}. \quad (62)$$

This dichotomy is unconditional, but its favorable alternative is not. A proof requiring the represented high face cannot discard either the base mass or the unrepresented high mass. In particular, the possibility $\mathcal{D}_{\text{miss}} > 0$ is exactly why the largest-prime census may vanish while the full Walsh diagonal is positive.

4.4 Why coherent energy is not a dyadic measure

Let $q(u) = (i, s(u))$ be the residual output key and define

$$A_k(i, s) := \sum_{\substack{u \in \mathcal{B}_k(y) \\ q(u) = (i, s)}} x_u. \quad (63)$$

The complete very-high residual vector is $A_H = \sum_k A_k$, and therefore

$$\mathcal{G}_H = \left\| \sum_k A_k \right\|_2^2, \quad (64)$$

not $\sum_k \|A_k\|_2^2$.

There is nevertheless one exact locality statement. If two atoms have the same residual key, then

$$d_m r = d_{m'} r', \quad d_m, d_{m'} \in [D, 2D], \quad \frac{1}{2} \leq \frac{r'}{r} \leq 2. \quad (65)$$

Thus

$$\langle A_k, A_\ell \rangle = 0 \quad (|k - \ell| \geq 2), \quad (66)$$

and

$$\boxed{\mathcal{G}_H = \sum_k \|A_k\|_2^2 + 2 \operatorname{Re} \sum_k \langle A_k, A_{k+1} \rangle.} \quad (67)$$

Adjacent blocks can still cancel exactly. After the native erasure $(i, s) \mapsto i$, even this residual-label locality is lost because different s values share the same native output. Hence the atomic pigeonhole theorem cannot be applied to a coherent norm without a new lower-reassembly estimate.

5 The selected cofactor core and the undeleted singleton bridge

The dyadic theorem localizes raw terminal energy, not a coherent output norm. We now connect one selected cell to the complete undeleted residual vector by the full-fiber singleton bridge of TPC-57. The distinction between a very-high root and its full terminal fiber is essential: a collision partner of a root with $p > y$ need not itself remain on the very-high face [5].

5.1 Core energy and full active multiplicity

Let

$$\mathcal{U}_{h_0}^{\text{phys}}(\zeta) := \{u \in \mathcal{U}_{h_0}^{\text{term}} : x_u \neq 0\}, \quad q(u) = (i(u), s(u)), \quad (68)$$

and define the complete active degree of a residual fiber by

$$n_a^{\text{phys}}(\zeta) := |q^{-1}(a) \cap \mathcal{U}_{h_0}^{\text{phys}}(\zeta)|. \quad (69)$$

The degree in (69) is computed in the whole terminal vector, without imposing $p > y$ or a cofactor block. The complete undeleted residual energy is

$$\mathcal{G}_{\text{full}}(\zeta) := \sum_a \left| \sum_{\substack{u \in \mathcal{U}_{h_0}^{\text{term}} \\ q(u)=a}} x_u \right|^2. \quad (70)$$

The high-only residual energy used by the TPC-58 high-column interface is

$$\mathcal{G}_H(y; \zeta) := \sum_a \left| \sum_{\substack{u \in \mathcal{U}_{h_0}^{\text{term}} \\ q(u)=a \\ p(u) > y}} x_u \right|^2. \quad (71)$$

This is the same quantity as $\mathcal{G}_H$ in (64). The full degree below is deliberately stronger than the active degree inside the high-only vector.

For a half-open cofactor core

$$\mathcal{B}_R(y) := \{u \in \mathcal{U}_{h_0, > y}^{\text{term}} : R \leq r_u < 2R\}, \quad (72)$$

put

$$\mathcal{D}_R(y; \zeta) := \sum_{u \in \mathcal{B}_R(y)} |x_u|^2, \quad (73)$$

$$\mathcal{I}_R^{\text{full}}(y; \zeta) := \sum_{u \in \mathcal{B}_R(y) \cap \mathcal{U}_{h_0}^{\text{phys}}(\zeta)} |x_u|^2 (n_{q(u)}^{\text{phys}}(\zeta) - 1). \quad (74)$$

Thus $\mathcal{I}_R^{\text{full}}$ is a rooted, coefficient-weighted incidence, not an unweighted pair count and not a multiplicity computed after restricting the vector to $\mathcal{B}_R(y)$.

Theorem 5.1 (Very-high core to high-only and full-output singleton bridge). *For every fixed $h_0 \neq 0$, every admissible cutoff y , every cofactor scale R , and every distinguished coefficient vector,*

$$\boxed{\min\{\mathcal{G}_H(y; \zeta), \mathcal{G}_{\text{full}}(\zeta)\} \geq (\mathcal{D}_R(y; \zeta) - \mathcal{I}_R^{\text{full}}(y; \zeta))_+}. \quad (75)$$

Consequently, if $\mathcal{D}_R(y; \zeta) > 0$ and

$$\mathcal{I}_R^{\text{full}}(y; \zeta) \leq (1 - \delta_R) \mathcal{D}_R(y; \zeta), \quad 0 < \delta_R \leq 1, \quad (76)$$

then

$$\boxed{\min\{\mathcal{G}_H(y; \zeta), \mathcal{G}_{\text{full}}(\zeta)\} \geq \delta_R \mathcal{D}_R(y; \zeta)}. \quad (77)$$

If $R = 2^{k^*}$ is the maximal-energy cell selected in [Theorem 4.2](#), then

$$\min\{\mathcal{G}_H(y; \zeta), \mathcal{G}_{\text{full}}(\zeta)\} \geq \frac{\delta_R}{B_X(y)} \mathcal{D}_H(y; \zeta) \quad (78)$$

whenever (76) holds.

Proof. If $n_a^{\text{phys}} = 1$, the unique active atom in the full fiber a survives with exactly its atomic energy both in (70) and, when it is a very-high root, in (71). Hence each of $\mathcal{G}_H$ and $\mathcal{G}_{\text{full}}$ dominates the energy of all core roots whose full active degree is one. A core root with full degree at least two contributes at most $|x_u|^2 (n_{q(u)}^{\text{phys}} - 1)$ to the energy removed from this singleton subcollection. Summing over core roots proves $\min\{\mathcal{G}_H, \mathcal{G}_{\text{full}}\} \geq \mathcal{D}_R - \mathcal{I}_R^{\text{full}}$; nonnegativity gives (75). Equation (77) follows from (76). For the selected dyadic cell, combine it with (54). $\square$

No comparison between the restricted grouped norm

$$\sum_a \left| \sum_{u \in q^{-1}(a) \cap \mathcal{B}_R(y)} x_u \right|^2$$

and either $\mathcal{G}_H(y; \zeta)$ or $\mathcal{G}_{\text{full}}(\zeta)$ occurs in the proof. Restoring omitted coordinates can cancel a restricted coherent sum. The theorem instead retains only roots that are singletons in the complete active fiber system. Those roots survive in particular in the high-only residual vector needed by the downstream native-erasure map.

5.2 Factor-exchange locality of every full partner

Define the enlarged, unrestricted terminal neighborhood

$$\mathcal{B}_R^+ := \{v \in \mathcal{U}_{h_0}^{\text{term}} : R/2 \leq r_v < 4R\}. \quad (79)$$

Unlike $\mathcal{B}_R(y)$, this set does not impose $p_v > y$.

Proposition 5.2 (Every full partner is cofactor-local). *If $u \in \mathcal{B}_R(y)$ and $v \in \mathcal{U}_{h_0}^{\text{term}}$ satisfies $q(v) = q(u)$, then*

$$\boxed{R/2 \leq r_v < 4R.} \quad (80)$$

Consequently, for every physical core root,

$$n_{q(u)}^{\text{phys}}(\zeta) = |q^{-1}(q(u)) \cap \mathcal{U}_{h_0}^{\text{phys}}(\zeta) \cap \mathcal{B}_R^+|, \quad (81)$$

and the exact incidence is

$$\boxed{\mathcal{I}_R^{\text{full}} = \sum_{\substack{u \in \mathcal{B}_R(y) \cap \mathcal{U}_{h_0}^{\text{phys}}(\zeta) \\ v \in \mathcal{B}_R^+ \cap \mathcal{U}_{h_0}^{\text{phys}}(\zeta) \\ v \neq u, q(v)=q(u)}} |x_u|^2.} \quad (82)$$

When R is dyadic, the enlarged range is the union of three adjacent half-open dyadic cells, so localization of all partners has $O(1)$ covering cost.

Proof. Write the two source rows as $m = d\ell$ and $m' = d'\ell'$. Equality of the residual keys gives

$$dr = d'r'. \quad (83)$$

Both source divisors belong to the same declared cell $[D, 2D]$. Therefore

$$r' = \frac{d}{d'}r, \quad \frac{1}{2} \leq \frac{d}{d'} \leq 2.$$

Together with $R \leq r < 2R$, this proves (80). Thus every full physical partner is already present in $\mathcal{B}_R^+$, proving (81); expanding $n_{q(u)}^{\text{phys}} - 1$ then gives (82). $\square$

The proposition is stronger than merely checking collisions inside the selected very-high cell, but weaker than an occupancy estimate. A carrier made entirely of active doubletons obeys the same $O(1)$ partner locality and still has no singleton energy. Thus bounded scale displacement does not make δ_R in (76) positive.

5.3 Critical compatibility does not create occupancy

At the critical cutoff $y = C_{\text{vh}}Q$, Theorem 2.3 gives

$$R_{\text{safe}}(y) \asymp R_{\text{max}}(y) \asymp J. \quad (84)$$

Hence the exponent $R \asymp J$ is once again compatible with the very-high dictionary. This removes the fixed positive-power support gap created by $y = X^{267/400+\eta}$; it proves none of the following:

- (i) that the fixed retained-pair carrier contains a cofactor in the critical window on every active row;
- (ii) that such a coordinate terminalizes or has nonzero physical energy;
- (iii) that the energy-rich dyadic cell k_* from Theorem 4.2 lies at scale J ; or
- (iv) that a positive weighted fraction of its full fibers are singletons.

The first two failures are measured by the occupancy and terminal minimax constants of Section 3; the fourth is precisely (76).

For comparison, retain the fixed-power cutoff

$$y = X^{267/400+\eta}, \quad \eta > 0. \quad (85)$$

Then every very-high atom satisfies

$$r \leq R_{\max}(y) = X^{133/400-\eta+o(1)}, \quad (86)$$

whereas an inherited root with $R \asymp J$ and every one of its factor-exchange partners lie in a constant-factor neighborhood of J . Thus, for large X ,

$$[R/2, 4R) \cap [1, R_{\max}(y)] = \emptyset. \quad (87)$$

The factor-exchange identity does not transport an old atom across this polynomial gap: it only locates other atoms already sharing the same full residual key. In particular, it cannot turn geometric incompatibility into cofactor occupancy or create terminal energy in a missing block.

5.4 The remaining arithmetic door

After high-face activation and dyadic localization, the exact remaining singleton target is

$$\boxed{\mathcal{I}_{R_*}^{\text{full}}(y; \zeta) \leq (1 - \delta_{R_*}) \mathcal{D}_{R_*}(y; \zeta), \quad \delta_{R_*} \geq X^{-\lambda_{\text{sing}}+o(1)},} \quad (88)$$

for the distinguished energy-rich block $R_* = 2^{k_*}$, or a stronger coefficient-uniform analogue on a fixed carrier. By (82), this is a local, weighted one-atom-versus-two-atom factor-exchange estimate retaining the literal terminal coefficient and the one physical shift h_0 . A support-only upper bound for the maximal fiber degree does not imply it.

The bridge theorem and partner localization are unconditional finite-dimensional and literal-support statements (L0/L1). Positivity or a polynomial lower bound for δ_{R_*} is a new arithmetic input; neither critical cutoff compatibility nor dyadic pigeonholing supplies it. Only a quantitative proof of (88), together with the upstream activation and downstream transfers, could contribute to an L2 advance. No undeleted asymptotic lower square, native-affine Gram transfer, parity improvement, or twin-prime consequence is claimed here.

6 Endpoint exponent audit and the revised loss ledger

The critical cutoff removes a geometrically forced positive-power displacement, but it does not improve the inherited determinant-ladder capacity or create singleton energy. This section performs the complete endpoint calculation for a general cutoff exponent and then replaces the single relocation symbol of TPC-58 by the two logically distinct gates identified above.

6.1 The general ξ exponent dictionary

Write

$$\omega_X = \frac{y}{Q} = X^{\xi+o(1)}, \quad 0 \leq \xi < \frac{133}{400}. \quad (89)$$

The upper restriction merely keeps the compatible cofactor ceiling polynomially growing; the identities below remain valid, with a bounded or empty cofactor range, at and beyond that endpoint. From (14) and Theorem 2.3, the largest compatible cofactor exponent is

$$\theta_c(\xi) := \frac{133}{400} - \xi. \quad (90)$$

The source-prime exponent and the high-cofactor singleton threshold exponent are, respectively,

$$\lambda_L := \frac{267}{400} - \frac{10049}{52500} = \frac{99979}{210000}, \quad (91)$$

$$\lambda_{LD^2} := \lambda_L + 2 \frac{10049}{52500} = \frac{180371}{210000}. \quad (92)$$

For later reference, a compatible very-high residual label $s = d_m r$ obeys

$$s \leq X^{\lambda_D + \theta_c(\xi) + o(1)} = X^{110021/210000 - \xi + o(1)}, \quad \lambda_D := \frac{10049}{52500}. \quad (93)$$

Thus every residual label remains of polynomial size, as required by the TPC-55/57 divisor bounds [4, 5]. Let $M_X(R)$ denote the corresponding coefficient-blind maximum number of terminal atoms in one residual fiber on a block $r \asymp R$.

Theorem 6.1 (Endpoint audit on every compatible cofactor block). *Let $R = X^{\theta_R + o(1)}$ be a dyadic cofactor block contained in the very-high compatible range, so that*

$$0 \leq \theta_R \leq \theta_c(\xi). \quad (94)$$

Then the following statements hold.

- (i) *The source-prime/cofactor ratio satisfies*

$$\frac{R}{L} \leq X^{-15077/105000 - \xi + o(1)}. \quad (95)$$

In particular $R = o(L)$, uniformly over the compatible range.

- (ii) *The exponent supplied by the coefficient-blind determinant-ladder capacity certificate is*

$$\mu_{\text{cap}}(R) := (\lambda_L - \theta_R)_+ = \lambda_L - \theta_R, \quad (96)$$

and therefore

$$\boxed{\mu_{\text{cap}}(R) \geq \frac{15077}{105000} + \xi.} \quad (97)$$

Its gap above the nominal endpoint allowance is

$$\boxed{\mu_{\text{cap}}(R) - \frac{1}{400} \geq \frac{29629}{210000} + \xi.} \quad (98)$$

- (iii) *The TPC-57 same-divisor exclusion condition $R > HL$, where $H = \text{rad } |h_0| = X^{o(1)}$, fails throughout the compatible range. The corresponding large-exchanged-factor lower bound is therefore exponent-trivial there.*

- (iv) *The automatic full-fiber singleton condition*

$$R > 48HLD^2 \quad (99)$$

is separated from the largest compatible block by the fixed power

$$\boxed{\lambda_{LD^2} - \theta_c(\xi) = \frac{55273}{105000} + \xi.} \quad (100)$$

Consequently the automatic-singleton range is empty on the present very-high carrier for every $\xi \geq 0$.

Proof. The right side of (94) and (91) give

$$\theta_R - \lambda_L \leq \frac{133}{400} - \xi - \frac{99979}{210000} = -\frac{15077}{105000} - \xi,$$

which proves (95). Since the right side is strictly negative, $(\lambda_L - \theta_R)_+ = \lambda_L - \theta_R$, and its minimum occurs at $\theta_R = \theta_c(\xi)$. Hence

$$\lambda_L - \theta_R \geq \frac{99979}{210000} - \left(\frac{133}{400} - \xi \right) = \frac{15077}{105000} + \xi.$$

Subtracting $1/400 = 525/210000$ proves (98). Because $H = X^{o(1)}$, (95) rules out $R > HL$. Finally,

$$\frac{180371}{210000} - \left(\frac{133}{400} - \xi \right) = \frac{55273}{105000} + \xi,$$

which proves the last assertion. Fixed constants such as $48H$ are absorbed by the displayed $X^{o(1)}$ term. $\square$

The capacity calculation is a statement about the strength of an *upper* certificate:

$$M_X(R) \ll X^{o(1)} \left(1 + \frac{L}{R} \right). \quad (101)$$

It does not assert that the actual multiplicity is as large as the right side. More importantly, even a degree-two carrier may have zero singleton energy. Thus (98) kills a support-only transversal closure at the $1/400$ endpoint; it is not a value that may be entered as a proved singleton-occupancy loss. A weighted distinguished-vector estimate can still be strictly better, but none is proved here.

Corollary 6.2 (Critical cutoff is support-optimal). *Under the declared information $\Delta_M = O(Q)$, and at the level of fixed-power exponents, among the certified polynomial schedules $\omega_X = X^{\xi+o(1)}$, $\xi \geq 0$, the critical choice*

$$\boxed{\xi = 0, \quad y \asymp Q} \quad (102)$$

maximizes the compatible cofactor exponent and simultaneously minimizes the certified capacity exponent, the failure exponent in R/L , and the gap to the automatic-singleton threshold. Every fixed $\xi > 0$ worsens all four quantities by exactly ξ .

This is an endpoint optimization only. It proves neither that a critical block is present in every active row nor that any such block terminalizes. Sharper support information such as $\Delta_M = o(Q)$ could permit a different sub- Q cutoff analysis; no such information is assumed here. Those questions are measured exactly by $\beta_{\mathcal{R}}$, $\rho_{\mathcal{R}}(\zeta)$, and $\tau_{\mathcal{R}}$ in Section 3.

6.2 Retiring relocation: feasibility before occupancy

TPC-58 used one symbol λ_{reloc} for the passage from a parent carrier to the smaller cofactor range produced by its fixed- η cutoff [6]. That notation mixed a support feasibility question with an energy-transfer estimate. They are not additive loss exponents. The correct replacement is the ordered pair

$$\boxed{\lambda_{\text{reloc}} \rightsquigarrow \{\text{cutoff feasibility prerequisite}\} \text{ followed by } \lambda_{\text{occ}}.} \quad (103)$$

The feasibility prerequisite is the trichotomy in Theorem 2.3: a candidate block is safe, lies in the bounded boundary region, or is empty. It has no associated lower-energy inequality. At the critical cutoff, blocks at exponent $133/400$ are feasible, so there is no separate geometric exponent to enter in an energy ledger. For the fixed- η cutoff, by contrast, the old $R \asymp J$ block is eventually empty; this is a closed gate, not an energy loss equal to η .

Once feasibility is established, the only parent-to-window energy charge is the actual occupancy exponent. A typical distinguished input would be

$$\mathcal{E}_{\mathcal{R}}(\zeta) \geq X^{-\lambda_{\text{occ}}+o(1)} \mathcal{E}_{\text{par}}(\zeta), \quad (104)$$

or the corresponding coefficient-uniform assertion $\beta_{\mathcal{R}} \geq X^{-\lambda_{\text{occ}}+o(1)}$. Neither follows from cutoff admissibility.

There is no hidden transport map behind (104). It is an orthogonal coordinate projection onto cofactor labels already present in the parent carrier. Moreover, factor exchange inside one residual fiber satisfies $r'/r \in [1/2, 2]$. It cannot move energy across the fixed-power gap created by a cutoff with $\xi > 0$.

6.3 The complete prospective endpoint ledger

Let $\mathcal{D}_{\text{up}}$ denote an explicitly declared upstream raw atomic energy. If a packet-selection theorem is used, its normalization is

$$\mathcal{E}_{\text{par}}(\zeta) \geq X^{-\lambda_{\text{sel}}+o(1)} \mathcal{D}_{\text{up}}, \quad (105)$$

where $\mathcal{E}_{\text{par}}$ is exactly the one-side slice energy defined in (27). For one distinguished coefficient vector, write the remaining conditional chain before inserting its exponents as

$$\begin{aligned} \mathcal{D}_{\text{up}} &\xrightarrow{\text{selection}} \mathcal{E}_{\text{par}} \xrightarrow{\text{occupancy}} \mathcal{E}_{\mathcal{R}} \\ &\xrightarrow{\text{terminal}} \mathcal{D}_{\mathcal{R}}^{\text{term}} \hookrightarrow \mathcal{D}_H \xrightarrow{\text{dyadic}} \mathcal{D}_{k_*} \\ &\xrightarrow{\text{singleton}} \mathcal{G}_H(y; \zeta) \xrightarrow{\text{native erasure}} \mathcal{N}_H \xrightarrow{\text{affine}} \mathcal{N}_{\text{aff}}^{\text{rep}} \\ &\xrightarrow{\text{off-carrier reassembly}} \mathcal{N}_{\text{slice}}^{\text{full}} \\ &\xrightarrow{\text{boundary/tail}} \text{parent arithmetic target.} \end{aligned} \quad (106)$$

Every arrow denotes a different assertion. The unlabeled inclusion is exact because the indicator defining $\mathcal{D}_{\mathcal{R}}^{\text{term}}$ already imposes $p > y$. Terminal activation includes primality, the shaped terminal cutoff, squarefree and coprimality conditions; it is not part of the cofactor projection. The high-only square $\mathcal{G}_H$, not the terminal square containing $p \leq y$, is the input to the TPC-58 native-erasure map.

Here the TPC-58 maps T_H and $C_{\mu, H}$ are understood on the same represented terminal/Walsh baseline $\text{Ran } \iota$ used to define $\mathcal{G}_H$. Thus $\mathcal{N}_{\text{aff}}^{\text{rep}}$ contains the base column and the represented high column, but not the missing-high coordinates measured by $\mathcal{D}_{\text{miss}}$ in (59). Restoring those coordinates is the displayed off-carrier reassembly arrow. It is a genuine lower-transfer gate: an omitted high vector can cancel the represented affine vector after restoration, so no norm monotonicity is available. If a future lower transfer proves

$$\mathcal{N}_{\text{slice}}^{\text{full}} \geq X^{-\lambda_{\text{reasm}}+o(1)} \mathcal{N}_{\text{aff}}^{\text{rep}}, \quad (107)$$

then λ_{reasm} is its separate ledger charge.

The least maximizing dyadic index $k_* = k_*(\zeta)$ is allowed to depend on this distinguished vector. Thus the zero block exponent in this chain is not a coefficient-uniform theorem for one fixed dyadic cell. Such a uniform route would have to fix the cell independently of ζ or prove a separate finite-cover lower reassembly.

If future work proves polynomial lower bounds for all arrows in (106), the complete exponent is

$$\boxed{\Lambda_{\text{tot}} := \lambda_{\text{sel}} + \lambda_{\text{occ}} + \lambda_{\text{blk}} + \lambda_{\text{term}} + \lambda_{\text{sing}} + \lambda_H + \lambda_{\text{aff}} + \lambda_{\text{reasm}} + \lambda_{\text{bdry}} + \lambda_{\text{tail}}.} \quad (108)$$

The critical cutoff establishes feasibility at exponent $R \asymp J$ without creating an energy-ledger entry. The one exact zero fixed-power energy-localization cost established here is

$$\boxed{\lambda_{\text{blk}} = 0,} \quad (109)$$

as a distinguished-vector statement relative to an already activated nonzero very-high terminal diagonal, as in [Theorem 4.2](#). No positive lower scale or numerical loss exponent is proved for the remaining arithmetic gates.

The safe endpoint requirement is the strict inequality

$$\boxed{\Lambda_{\text{tot}} < \frac{1}{400}}. \quad (110)$$

Equality is not accepted without a constant-level comparison and quantified remainders. In particular, the logarithmic factor in the dyadic pigeonhole is $X^{o(1)}$ and has zero fixed exponent, but it can still spoil a closure attempted exactly at $1/400$.

6.4 Level and kill-test boundary

The conclusions of this section have the following precise status.

Exact L0/L1 progress.

The rational endpoint identities, cutoff optimization, support-capacity audit, empty automatic-singleton range, feasibility/occupancy split, and strict loss ledger are exact. Their use of $s = d_m r$, the literal fixed shift, and raw cofactor atoms is the L1 specialization.

Closed support-only doors.

On every compatible block, the available transversal/capacity exponent already exceeds $1/400$, while the deterministic same-divisor and automatic-singleton thresholds are outside the carrier. Increasing ξ cannot repair either failure.

Open arithmetic doors.

A positive distinguished or uniform cofactor occupancy, terminal activation, weighted singleton fraction, native-erasure angle, represented affine noncancellation angle, missing-high/off-carrier reassembly, and boundary/tail reconnection remain independent hypotheses.

Not proved (L2).

No positive fixed- h_0 arithmetic lower bound, parity improvement, prime-pair asymptotic, or twin-prime conclusion follows from the endpoint optimization.

Thus the critical cutoff genuinely removes one artificial positive-power support displacement and the dyadic theorem removes a second bookkeeping loss. The remaining obstruction is no longer cutoff geometry: it is the actual coefficient-weighted occupancy and activation of the compatible carrier, followed by the two undeleted grouping gates and restoration of the unrepresented high coordinates.

7 Six-point audit, claim boundary, and the next arithmetic doors

The critical cutoff changes the route in one precise way. It replaces the fixed-power displacement of the TPC-58 very-high face by an admissible support choice at the same exponent as the inherited cofactor packet:

$$y = C_{\text{vh}} Q, \quad R_{\text{safe}}(y) \asymp R_{\text{max}}(y) \asymp J. \quad (111)$$

Together with the exact dyadic partition, this proves one support feasibility statement and one zero fixed-power energy-localization cost:

$$\text{the cutoff gate is open at } R \asymp J, \quad \lambda_{\text{blk}} = 0 \quad \text{after } \mathcal{D}_H(y; \zeta) > 0. \quad (112)$$

Neither statement is an occupancy or primality theorem. Cutoff feasibility is not itself an additive energy exponent. The constants $\beta_{\mathcal{R}}$, $\rho_{\mathcal{R}}(\zeta)$, and $\tau_{\mathcal{R}}$ remain the first quantitative gates, and the undeleted singleton, native, and affine gates remain downstream of them. Restoring high Walsh coordinates outside the represented terminal image is a further nonmonotone reassembly gate.

7.1 Six-point physical audit

We record the six checks that must be passed before an exact support or linear-algebra statement is promoted to progress on the physical endpoint route. Passing these checks identifies the correct object; it does not prove that any of the open lower constants is positive.

(A1) *Literal physical coefficients.* The pre-terminal and terminal atoms are, respectively,

$$[G_X^L]_{m,r,j,\gamma} = c_{m,j,\gamma} \zeta_m, \quad x_u = \mu(r_u) \Lambda_X(mj + h_0) c_{m,j,\gamma} \zeta_m. \quad (113)$$

The coefficient $c_{m,j,\gamma}$ retains the inherited source shape, Möbius and prime-logarithm factors, masks, opposite-row factor, orbit phase, and four-factor profile. The terminal multiplier is applied only at terminalization. No abstract sign family, coefficient-blind support vector, or renormalized terminal atom is substituted for these physical coefficients. The terminal/Walsh comparison uses the calibrated common gauge of TPC-58, so no second Möbius factor is inserted.

(A2) *One fixed physical shift.* Every terminal coordinate obeys

$$mj + h_0 = p_u r_u \quad (114)$$

for the one declared nonzero integer h_0 . The cutoff theorem, largest-prime census, full-fiber collision relation, and endpoint audit are pointwise at this shift. There is no average over shifts, intercepts, or frozen parameter families.

(A3) *Physical atomic normalization.* The parent energy $\mathcal{E}_{\text{par}}$, compatible-window energy $\mathcal{E}_{\mathcal{R}}$, terminal diagonal $\mathcal{D}_H$, and selected-block diagonal $\mathcal{D}_R$ all use raw counting measure on their declared atoms. Repeated cofactor values in different coordinates remain distinct orthogonal atoms until the stated grouping. In particular, the shaped weight Λ_X is not moved into a new input norm, and the atomic dyadic identity is not asserted for a coherent grouped square.

(A4) *Canonical representation and minimality scope.* The baseline retained-pair carrier $\mathcal{I}_X$, orbit cell, opposite-row cell, and cutoff are fixed independently of ζ . On the very-high face, $p_u > y > \sqrt{N_+}$ gives a unique largest-prime representation and hence a canonical cofactor r_u . The projection onto a fixed set $\mathcal{R}$ is therefore an actual coordinate projection, not a transport between cofactor labels. By contrast, the maximal dyadic block $k_*(\zeta)$ may depend on the distinguished physical vector. Its pigeonhole estimate is a distinguished-vector statement and is not silently promoted to a coefficient-uniform fixed-block theorem. The uniqueness assertion is canonicity of the very-high atomic representation on the declared one-side baseline; it is not a claim that the full coefficient parametrization is globally dimension-minimal.

(A5) *Actual active support.* Cutoff compatibility, cofactor occupancy, and terminal activation are kept separate. Equation (111) says only that exponent-133/400 cofactors are allowed by the literal very-high inequalities. The exact tests

$$\beta_{\mathcal{R}} = \min_{m: B_m N(m) > 0} \frac{N_{\mathcal{R}}(m)}{N(m)}, \quad \tau_{\mathcal{R}} = \min_{m: B_m N_{\mathcal{R}}(m) > 0} \frac{T_{\mathcal{R}}(m)}{B_m N_{\mathcal{R}}(m)} \quad (115)$$

may still vanish. Moreover, the degree used in the singleton bridge is the degree in the complete active terminal vector. Collision partners with $p \leq y$ are not deleted merely because the root lies on the very-high face. The decomposition (60) also keeps the missing high Walsh mass $\mathcal{D}_{\text{miss}}$ separate from the represented terminal image. The TPC-58 high map used downstream is restricted to that represented image; putting $\mathcal{D}_{\text{miss}}$ back requires its own reassembly estimate.

- (A6) *Strict endpoint budget.* Cutoff feasibility is audited before the energy ledger. Actual occupancy, terminal activation, singleton survival, the native Gram, represented affine completion, missing-high reassembly, and downstream boundary/tail transfers are separate ledger entries. Only the conditional localization cost λ_{blk} is proved zero here. Every route to the endpoint must satisfy

$$\Lambda_{\text{tot}} < \frac{1}{400} \tag{116}$$

with Λ_{tot} as in (108). Equality is not accepted without closed leading constants and quantified remainders. The coefficient-blind capacity exponent in (98) exceeds the allowance; it closes that support-only certificate and is not entered as a proved physical singleton loss.

7.2 Claim table

Statement	Status
Critical admissible cutoff $y = C_{\text{vh}}Q$	Exact support theorem (L0), specialized to the literal fixed- h_0 packet without changing its coefficients or normalization (L1).
Cutoff-cofactor reciprocity	Exact relation $R_{\text{safe}} \asymp R_{\text{max}} \asymp J/\omega_X$ and exact comparison of the critical, subpower, and fixed-power regimes (L0/L1). It is not an occupancy estimate.
Sharp cofactor-occupancy constant	Exact rowwise minimax formula for $\beta_{\mathcal{R}}$, and exact distinguished ratio $\rho_{\mathcal{R}}(\zeta)$ (L0). Positivity or a useful endpoint exponent for either quantity is not proved.
Terminal-activation constant	Exact rowwise minimax formula for $\tau_{\mathcal{R}}$ and combined constant $\chi_{\mathcal{R}}$ (L0), using the literal terminal weights (L1). Positivity is not proved.
Largest-prime census	Exact characterization of every very-high terminal atom by $P^+(mj + h_0)$, including all retained-pair and terminal masks at one fixed shift (L1). It is an identity, not a density theorem.
Dyadic localization	Exact raw-energy partition and an $X^{-o(1)}$ maximal-block loss after $\mathcal{D}_H > 0$ (L0/L1). Thus $\lambda_{\text{blk}} = 0$ conditionally; no parent-to-high-face transfer is supplied.
Base/represented/missing decomposition	Exact atomic decomposition including high Walsh mass absent from the retained terminal image (L0/L1). The alternative favorable to a proof using the represented high face is not forced.
Core-to-high/full singleton bridge	Exact lower floor for both the high-only residual square used downstream and the complete undeleted residual square, using roots singleton in the complete terminal fibers; every collision partner is localized to a constant-scale neighborhood (L1). A positive weighted singleton fraction is not proved.
Critical endpoint support audit	Exact proof that the cutoff feasibility gate is open at $R \asymp J$; every fixed-power increase in the cutoff lowers the compatible cofactor exponent by the same power (L1 route correction). Feasibility is not an additive energy loss.
Support-only singleton closures	The inherited capacity bound already exceeds the 1/400 allowance, and the same-divisor and automatic-singleton thresholds lie outside the compatible carrier (L1 no-go for those specific certificates). This is not a lower bound for the actual collision energy.
Native and affine transfers	The exact downstream constants α_H and δ_{aff} remain those of TPC-58 on the represented terminal/Walsh baseline. Their positivity and scale-uniform size are not consequences of critical cutoff tuning.
Missing-high reassembly	Restoring $\mathcal{W}_H \setminus \text{Ran } \iota$ after the represented affine transfer is a separate nonmonotone gate. No positive reassembly constant or exponent is proved.
Endpoint conclusion	Conditional exponent ledger only. No positive fixed- h_0 arithmetic lower square, parity improvement, prime-pair asymptotic, or twin-prime conclusion is proved (no L2 theorem).

Here L0 denotes exact finite-dimensional identities, minimax laws, support geometry, and sharp route-specific obstructions. L1 denotes their exact specialization to the inherited literal coefficients, raw atomic norm, actual carrier, and one fixed nonzero shift. An L2 result would require a new quantitative arithmetic estimate for the distinguished physical vector or the full canonical coefficient image, strong enough to survive every arrow in (106). No result in this paper is classified as L2.

7.3 The next seven doors and their kill tests

The next work should follow the transfer order. A downstream Gram calculation cannot repair an empty compatible face, and a favorable atomic block cannot be returned to the undeleted coherent vector by monotonicity.

Door A. *Critical-window occupancy:* $\beta_{\mathcal{R}}$ or $\rho_{\mathcal{R}}(\zeta)$. Fix a coefficient-independent critical cofactor set $\mathcal{R}_{\text{crit}} \subset [c_-, c_+]$ allowed by [Theorems 2.1](#) and [2.3](#). The uniform target is

$$\beta_{\mathcal{R}_{\text{crit}}} \geq X^{-\lambda_{\text{occ}} + o(1)}, \quad (117)$$

while the weaker distinguished target is

$$\rho_{\mathcal{R}_{\text{crit}}}(\zeta) \geq X^{-\lambda_{\text{occ}} + o(1)}. \quad (118)$$

These require information about the cofactors already present in the literal retained-pair carrier; there is no relocation map to analyze. The exact uniform kill test is an active row with $N_{\mathcal{R}_{\text{crit}}}(m) = 0$, which forces $\beta = 0$. A distinguished route survives that failure only if its physical coefficient vanishes on every offending row and its own ratio remains quantitatively positive. The adaptive block $k_*(\zeta)$ may be used only after this distinction is stated.

Door B. *Terminal activation:* $\tau_{\mathcal{R}}$. On a surviving compatible set, prove

$$\tau_{\mathcal{R}} \geq X^{-\lambda_{\text{term}} + o(1)}, \quad (119)$$

or the explicitly labeled distinguished-vector analogue comparing $\mathcal{D}_H(y; \zeta)$ with $\mathcal{E}_{\mathcal{R}}(\zeta)$. The largest-prime census reduces the problem to the fixed-shift values $P^+(mj + h_0)$, but does not estimate their density. Every squarefree, coprimality, support, and shaped-terminal factor must stay in $T_{\mathcal{R}}(m)$. An active compatible row with $T_{\mathcal{R}}(m) = 0$ kills the coefficient-uniform route. If the best activation loss consumes the remaining endpoint slack, the route stops before dyadic or Gram analysis.

Door C. *Weighted full-fiber singleton collision.* After $\mathcal{D}_H > 0$, select the maximal atomic block and prove the literal one-sided estimate

$$\mathcal{I}_{R_*}^{\text{full}}(y; \zeta) \leq (1 - \delta_{R_*})\mathcal{D}_{R_*}(y; \zeta), \quad \delta_{R_*} \geq X^{-\lambda_{\text{sing}} + o(1)}. \quad (120)$$

The degree is computed in the complete terminal fiber, and every partner in the enlarged block $[R_*/2, 4R_*]$ is retained whether or not its prime exceeds y . Unweighted degree bounds, the coefficient-blind determinant capacity, and singleton counts inside the selected face alone do not prove this inequality. If $\delta_{R_*} = 0$, or if every valid lower exponent already exceeds the remaining budget, the present singleton route closes.

Door D. *Native residual-label erasure:* α_H . Only after the undeleted residual energy has been obtained should one form the nonzero actual image $E_H = \text{Ran } T_H$ and seek

$$\alpha_H \geq X^{-\lambda_H + o(1)}. \quad (121)$$

The exact uniform kill test remains

$$E_H \cap \text{Ker } C_{\mu, H} \neq \{0\}. \quad (122)$$

A positive singleton fraction controls the first grouping $p \mapsto (i, s)$; it does not control the second grouping $(i, s) \mapsto i$. A finite-scale numerical eigenvalue is scouting evidence, not a scale-uniform lower certificate.

Door E. *Base-high affine transversality:* δ_{aff} . Conditional on the native gate, form the nonzero represented feasible pair image, retain the joint dependence of the base and represented high columns on the same coefficient vector, and prove

$$\delta_{\text{aff}} \geq X^{-\lambda_{\text{aff}}+o(1)}. \quad (123)$$

The exact uniform kill test is a nonzero intersection of the actual feasible pair image with the anti-diagonal,

$$\mathcal{F}_{\text{aff}} \cap \{(g, -g) : g \in \mathcal{H}_{\text{out}}\} \neq \{0\}. \quad (124)$$

Treating the base and high vectors as independent, or replacing the physical coefficient vector by a favorable adaptive pair, changes the problem. If the uniform image meets the anti-diagonal, only an explicitly distinguished-vector route may remain.

Door F. *Missing-high/off-carrier reassembly.* Let $g_{\text{rep}}(\zeta)$ be the represented affine output after Door E and let $h_{\text{miss}}(\zeta)$ be the native contribution of the high Walsh coordinates in $\mathcal{W}_H(y) \setminus \text{Ran } \iota$. Both are driven by the same row coefficient vector. The required lower transfer has the form

$$\|g_{\text{rep}}(\zeta) + h_{\text{miss}}(\zeta)\|_2^2 \geq X^{-\lambda_{\text{reasm}}+o(1)} \|g_{\text{rep}}(\zeta)\|_2^2. \quad (125)$$

Its coefficient-uniform zero test is a nonzero actual pair $(g_{\text{rep}}, h_{\text{miss}}) = (g, -g)$. The scalar $\mathcal{D}_{\text{miss}}$ records omitted atomic mass but cannot rule out such vector cancellation. If the anti-diagonal is met, a distinguished-vector estimate or a different carrier is required; simply restoring the omitted coordinates is not norm-monotone.

Door G. *Strict endpoint budget and route decision.* Insert every proved exponent into

$$\begin{aligned} \Lambda_{\text{tot}} = & \lambda_{\text{sel}} + \lambda_{\text{occ}} + \lambda_{\text{blk}} + \lambda_{\text{term}} + \lambda_{\text{sing}} \\ & + \lambda_H + \lambda_{\text{aff}} + \lambda_{\text{reasm}} + \lambda_{\text{bdry}} + \lambda_{\text{tail}}, \end{aligned} \quad (126)$$

retaining $\lambda_{\text{blk}} = 0$ only with the distinguished-vector scope in (112); cutoff feasibility is checked before this sum. Continue the endpoint route only if $\Lambda_{\text{tot}} < 1/400$. At equality, stop unless leading constants and all remainders are closed. If a kernel or zero-occupancy test fires, or a necessary loss alone exhausts the slack, record the resulting route-specific obstruction rather than replacing the physical vector, shift, normalization, or carrier.

Conclusion. The critical-cutoff analysis is a genuine L1 route correction. It proves that the earlier positive-power cofactor displacement was caused by a nonoptimal cutoff schedule, splits geometry from actual occupancy, and shows that dyadic localization costs no positive power after the very-high face has been activated. The surviving route is narrower and auditable: critical-window occupancy, terminal activation, weighted singleton survival, native erasure, represented affine transversality, missing-high reassembly, and the strict endpoint ledger must be passed in that order. None is bypassed here. In particular, this paper contains no L2 parity improvement, prime-pair asymptotic, or twin-prime implication.

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